WHAT ARE VECTOR DATABASES?

Get Started

Table of contents

- What are vector databases?
- How do vector databases work?
- How does generative AI relate to vector databases?
- Why is vector search critical?
- Use cases for vector databases
- MongoDB Vector Search: A game-changer
- MongoDB Vector Search: For intelligent applications powered by semantic search
- FAQs

What are vector databases?

A vector database—also known as a vector search database or vector similarity search engine—stores, retrieves, and searches data represented as vectors in high-dimensional space. It enables efficient similarity searches by comparing vector embeddings rather than relying on exact term matches.

How does it work? What are some common use cases? And why is MongoDB Vector Search playing a significant role in the generative AI discussion?

To understand vector databases, you need to first understand the vector.

In math and physics, a vector is a quantity that has both magnitude (or size) and direction. A vector can be broken down into components. For example, in a two-dimensional space, a vector has an X (horizontal) and Y (vertical) component.

In data science and machine learning, a vector is an ordered list or sequence of numbers that represents data. A vector can represent any type of data, including unstructured data (or data without a pre-defined data model or schema)—from text to image, audio to video. A vector is usually represented as arrays or lists of numbers where each number in the list represents a specific feature or attribute of that data.

For example, imagine you have a large collection of cat photos. Each image is a piece of unstructured data. But you can represent each image as a vector by extracting features, such as the following:

- Average color
- Color histogram
- Texture histogram
- The presence or absence of ears, whiskers, and a tail

Vectorization is the process of converting data—such as words, images, or audio—into numerical vector embeddings, where each data point is represented in a high-dimensional space.

Instead of rows and columns typical of relational databases, vector databases represent data as points in a multi-dimensional space. Vector databases are ideal for applications that require rapid and accurate matching of data based on similarity rather than exact values.

If you're building generative AI applications, a vector database is tailored to efficiently process vast volumes of vectorized data, ensuring faster queries and processing speeds.

Vector databases provide native capabilities to manage and index vector embeddings. Some, including MongoDB Atlas, also offer distributed scaling across multiple nodes to handle large datasets efficiently.

How do vector databases work?

Central to the functionality of a vector database is the principle of embeddings. In essence, a vector or embedding model translates data into a consistent format: vectors.

While a vector is fundamentally an ordered list of numbers, an embedding is a type of vector generated by a machine learning model to represent complex data such as text, images, or audio in a shared numerical space.

In practice, a vector database handles vector embeddings generated by machine learning models or large language models, indexing them for rapid retrieval based on semantic meaning.

Transformations—the process of converting data from one format to another—situate vectors in multi-dimensional vector space. In this vector space, data points with similar attributes or characteristics are located close to one another, forming clusters that reflect shared features in the data.

Vector embeddings are not just numerical translations; they encapsulate the deeper semantic essence and the contextual nuances of the original data. This makes them invaluable assets for a range of AI applications—from natural language processing (NLP) to sentiment analysis to text categorization.

Querying a vector database is different from querying a conventional database. When a query vector embedding is compared against stored data, the database measures similarity using distance metrics such as cosine similarity, Euclidean distance, or dot product to return the most relevant results.

Instead of hunting for precise matches between identical vectors, a vector database uses similarity search to identify vectors that reside in close proximity to the given query vector, within the multi-dimensional space. This approach not only more closely aligns with the inherent nature of the data but also offers a speed and efficiency that traditional search can't match.

Words, sentences, and even entire documents can be transformed into vectors that capture their essence. For example, a standard word embedding method is Word2Vec. With Word2Vec, words with similar meanings are represented by vectors that are close in a multi-dimensional space.

A classic Word2Vec example illustrates how embeddings can encode relationships: “king – man + woman ≈ queen.” While this analogy demonstrates semantic relationships, newer transformer-based embeddings capture meaning in more complex, non-linear ways.

Even with their intricate patterns and colors, images can be translated into vectors. For instance, in a dataset teeming with animal images, a trained convolutional neural network (CNN) would cluster all dog images close together, distinctly separate from, say, clusters of cats or birds.

This process allows vector databases to efficiently store vectors and deliver contextually relevant matches even across high-dimensional data structures.

By capturing the inherent data structure, and patterns within the data, vector embeddings offer semantically enriched portrayals. This richness not only facilitates a deeper understanding of the data but also expedites computations related to determining relationships and gauging similarities between different entities.

How does generative AI relate to vector databases?

You've heard the hype about generative AI. Across the economy—from healthcare to finance, retail to government agencies—organizations are looking for ways to leverage it. It seems like every CEO wants to roll out applications as fast as possible.

It's more than just hype. Generative AI could infuse trillions into the global economy.

Central to this transformational technology is the mathematical concept of the vector. Through vectorization and the prowess of large language models (LLMs), generative AI achieves its game-changing potential. In the era of generative AI, vector embeddings lay the groundwork; vector databases amplify its impact. Unlike traditional databases, which rely on predefined schemas and exact matches, vector databases handle complex unstructured data through similarity search in multidimensional space.

Why is vector search critical?

Vector search is the capability that enables semantic and similarity-based retrieval across high-dimensional data. It’s now integrated into general-purpose databases such as MongoDB Atlas in addition to specialized vector databases. Instead of relying on exact keyword matches, it retrieves contextually similar results, making it essential for applications like recommendation systems and generative AI.

To explore how MongoDB powers these capabilities, visit the MongoDB Vector Search page.

Use cases for vector databases

The global economic landscape is complex and competitive—and data remains at its core. In the past, many have called data the “new oil.” In the generative AI era, vector embeddings are the oil and vector databases have emerged as sophisticated refineries, adept at processing high-dimensional data and executing similarity searches.

For the C-suite, generative AI isn't just a buzzword; it's a strategy. For developers, the primary allure of vector databases is efficiency. Traditional databases might require complex query structures to fetch relevant data, especially when dealing with vast datasets. Vector databases simplify this, allowing developers to retrieve data based on similarity, reducing both the complexity of the code and the time taken for data retrieval.

A sampling of vector database use cases

Image and video recognition: Visual content dominates our visual culture, and vector databases shine brightly in it. They are adept at sifting through vast repositories of images and videos to pinpoint those that bear a striking resemblance to a given input. This isn't just about matching pixel-by-pixel; it's about understanding the underlying patterns and features. Such capabilities are crucial for applications like facial recognition, object detection, and even copyright infringement detection in media platforms.

Natural language processing and text search: Synonyms, paraphrasing, and context can make exact text matching a daunting task. However, vector databases can discern the semantic essence of phrases or sentences, enabling them to identify matches that might not be identical in terms of wording but are contextually similar. This prowess is a game-changer for chatbots, ensuring they respond aptly to user queries. Similarly, search engines can deliver more relevant results, enhancing user experience.

Recommendation systems: Vector databases play a pivotal role in personalization. By understanding user preferences and analyzing patterns, these databases can suggest songs that resonate with a listener's taste or products that align with a shopper's preferences. It's all about gauging similarity and delivering content or products that strike a chord with the user.

Emerging applications: The horizon of vector databases is ever-expanding. In healthcare, they're aiding drug discovery by analyzing molecular structures for potential therapeutic properties. In the financial sector, vector databases are assisting in anomaly detection, spotting unusual patterns that might indicate fraudulent activities.

These use cases rely on vector databases’ ability to perform efficient similarity searches across high-dimensional vector data. In each scenario, a vector index organizes stored vectors to enable rapid nearest neighbor search, reducing query latency and improving scalability.

With the ascent of generative AI, vector databases emerge as vital enablers, assisting developers in transforming intricate AI blueprints into practical, value-driven tools.

MongoDB Vector Search: A game-changer

MongoDB Vector Search is the latest addition to MongoDB. MongoDB’s vector database capabilities let teams unify semantic, text, and structured data queries without cumbersome integration processes. It enables customers to build intelligent applications powered by semantic search and generative AI over any type of data. Visit the MongoDB Vector Search Quick Start guide and create your first index in minutes.

Historically, development teams seeking a vector database for tasks like image or efficient similarity search faced a dilemma: Opt for a bolt-on vector database, adding another tool to the tech stack, or juggle a mix of search tools and open-source solutions. Using a full-text search for semantic capabilities often meant developers were bogged down with extensive synonym mapping. The limitations were clear: If users weren't precise in their queries, the results were far from relevant.

Such challenges meant:

- An added system to oversee.
- The need for specialized skill sets.
- The mental strain of constantly updating synonym mappings.
- A subpar user experience for imprecise queries.
- Valuable engineering time diverted from core tasks.

MongoDB Vector Search simplifies designing applications enriched by semantic search and generative AI, capable of processing a range of data types, from videos to social media content. Harnessing the robustness of MongoDB Atlas, Vector Search allows developers to craft cutting-edge, relevance-based search tools on a trusted platform with a unified query interface.

Vector Search enables MongoDB Atlas to retrieve semantically relevant results without requiring predefined synonyms. Even when users don't know what they're looking for, Vector Search is able to return relevant results based on the meaning of the query. For example, a search for “ice cream” would return “sundae,” even if the user didn't know sundaes existed.

When you use Vector Search, you'll store vector embeddings alongside the original data and metadata in MongoDB Atlas. This ensures any updates or additions to your vector data are instantly synchronized, streamlining the architecture and offering a unified developer experience.

With Vector Search, you'll index and query data using approximate nearest neighbor (ANN) algorithms such as HNSW, which efficiently locate similar vectors in high-dimensional space. MongoDB supports multiple distance metrics and hybrid search, combining vector, text, and structured filters.

You can create vastly improved search experiences that address use cases that traditional search tools can't, including:

- Semantic search: This allows for context-driven searches. For instance, a search for “ice cream” might yield results like “sundae” without any pre-set synonyms.
- Enhanced recommendations: If a user searches for a lawn mower, the system might also suggest related lawn-care items.
- Diverse media searches: Whether it's hunting for images resonating with terms like “happy families” or sifting through audio logs for specific phrases, vector search is up to the task.
- Hybrid search: This combines the strengths of vector search with traditional full-text search, enriching the results.
- Long-term memory for LLMs: This provides proprietary business data context to large language models, refining their output accuracy—a foundational capability for AI agents that need to recall past interactions and act on current data.

MongoDB Vector Search is compatible with popular application frameworks like LlamaIndex and LangChain. It also integrates seamlessly with ecosystem partners such as Google Vertex AI, AWS, Azure, and Databricks, ensuring proprietary business data enhances the performance and accuracy of AI-powered applications.

MongoDB Vector Search: For intelligent applications powered by semantic search

Vector databases, with their unique approach to data storage and retrieval, are changing the way we think about databases. Their ability to perform rapid similarity searches makes them indispensable in today's data-driven world. And when combined with the power and flexibility of MongoDB Atlas, they offer a solution that's hard to beat.

MongoDB Vector Search powers advanced use cases—like semantic search, image search, and similarity search—that can't be addressed by traditional full-text search. Developers can store their vector embeddings in MongoDB, supplement their existing search functionality with machine learning models, and query them to get relevant, contextual results. Engineering leaders benefit from the peace of mind that comes with running Atlas: a fully managed, modern, multi-cloud database.

Whether you're building a recommendation system, a search engine, or any other application that requires fast and accurate data matching, consider leveraging the combined power of vector databases and MongoDB. The future is vectorized, and MongoDB is here to help you navigate it.

FAQs

What is MongoDB Vector Search?

MongoDB Vector Search enables customers to build intelligent applications powered by semantic search and generative AI over any type of data.

What is the approximate nearest neighbor search?

What is a vector index?

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